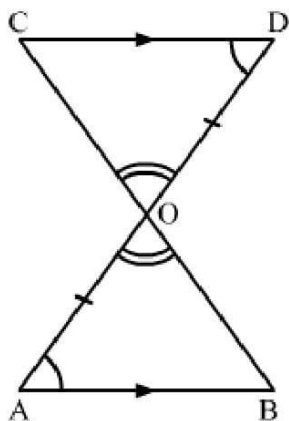


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EXERCISE 9A

Solution 1:



Given: $AB \parallel CD$ and O is the midpoint of AD.

To prove:

(i) $\triangle AOB \cong \triangle DOC$.

(ii) O is the midpoint of BC.

Proof:

(i) In $\triangle AOB$ and $\triangle DOC$,

$\angle BAO = \angle CDO$ (Alternate interior angles)

$AO = DO$ ($\because$ O is the midpoint of AD)

$\angle AOB = \angle DOC$ (Vertically opposite angles)

$\therefore$ By ASA congruence criteria, we get

$\triangle AOB \cong \triangle DOC$

(ii) We have,

$\triangle AOB \cong \triangle DOC$

$\therefore BO = CO$ (CPCT)

Hence, O is the midpoint of BC.

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Solution 2:

Given: AD and BC are equal perpendiculars to a line segment AB.

To prove: CD bisects AB

Proof:

In $\triangle AOD$ and $\triangle BOC$,

$\angle DAO = \angle CBO = 90^\circ$ (Given)

$AD = BC$ (Given)

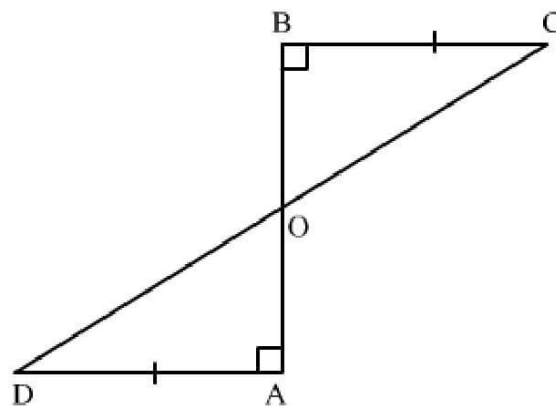
$\angle DOA = \angle COB$ (Vertically opposite angles)

$\therefore$ By AAS congruence criteria, we get

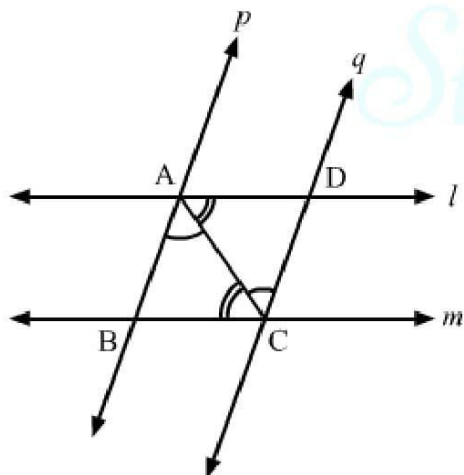
$\triangle AOD \cong \triangle BOC$

$\therefore AO = BO$ (CPCT)

Hence, CD bisects AB.



Solution 3:



Given: l and m are intersected by two parallel lines p and q.

To prove: $\triangle ABC \cong \triangle CDA$

Proof:

In $\triangle ABC$ and $\triangle CDA$,

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$\angle BAC = \angle DCA$	(Alternate angles)
$\angle BCA = \angle DAC$	(Alternate angles)
$AC = CA$	(Common side)

$\therefore$ By ASA congruence criterion, we get

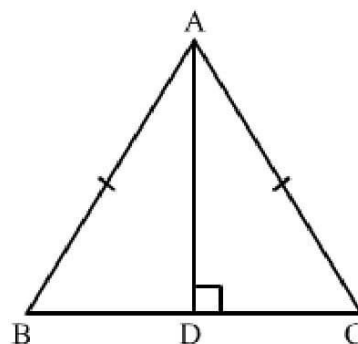
$$\triangle ABC \cong \triangle CDA$$

Solution 4:

Given: AD is an altitude of an isosceles $\triangle ABC$ in which $AB = AC$.

To prove:

- (i) AD bisects BC,
- (ii) AD bisects $\angle A$



Proof:

- (i) In $\triangle ABD$ and $\triangle ACD$,

$$\angle ADB = \angle ADC = 90^\circ \quad (AD \perp BC)$$

$$AB = AC \quad (\text{Given})$$

$$AD = AD \quad (\text{Common side})$$

$\therefore$ By RHS congruence criterion, we get

$$\triangle ABD \cong \triangle ACD$$

$$\therefore BD = CD \quad (\text{CPCT})$$

Hence, AD bisects BC.

- (ii) We have,

$$\triangle ABD \cong \triangle ACD$$

$$\therefore \angle BAD = \angle CAD \quad (\text{CPCT})$$

Hence, AD bisects $\angle A$.

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Solution 5:

In $\triangle ABE$ and $\triangle ACF$,

$$BE = CF \quad (\text{Given})$$

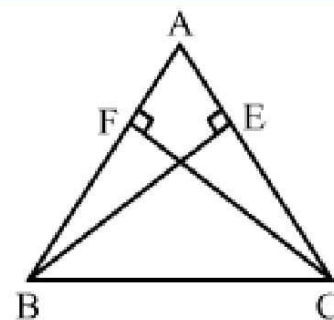
$$\angle BEA = \angle CFA = 90^\circ$$

$$\angle A = \angle A \quad (\text{Common})$$

$\therefore$ By AAS congruence criterion, we get

$$(i) \triangle ABE \cong \triangle ACF$$

$$(ii) AB = AC \quad (\text{CPCT})$$



Solution 6:

Given: $\triangle ABC$ and $\triangle DBC$ are two isosceles triangles on the same base BC.

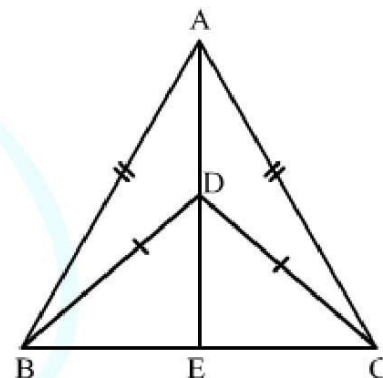
To prove:

$$(i) \triangle ABD \cong \triangle ACD$$

$$(ii) \triangle ABE \cong \triangle ACE$$

$$(iii) AE \text{ bisects } \angle A \text{ as well as } \angle D$$

$$(iv) AE \text{ is the perpendicular bisector of } BC$$



Proof:

$$(i) \text{ In } \triangle ABD \text{ and } \triangle ACD,$$

$$BD = CD \quad [\because \triangle DBC \text{ is an isosceles triangle}]$$

$$AB = AC \quad [\because \triangle ABC \text{ is an isosceles triangle}]$$

$$AD = AD \quad [\text{Common side}]$$

$\therefore$ By SSS congruence criterion, we get

$$\triangle ABD \cong \triangle ACD$$

$$\text{Also, } \angle BAD = \angle CAD \quad (\text{CPCT})$$

$$\text{or, } \angle BAE = \angle CAE \quad \dots (1)$$

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(ii)

In $\triangle ABE$ and $\triangle ACE$,

$$AB = AC \quad [\because \triangle ABC \text{ is an isosceles triangle}]$$

$$\angle BAE = \angle CAE \quad [\text{Proved above}]$$

$$AE = AE \quad [\text{Common side}]$$

$\therefore$ By SAS congruence criteria,

$$\triangle ABE \cong \triangle ACE$$

$$\text{Also, } BE = CE \quad \dots(2) \quad [\text{CPCT}]$$

$$\text{And, } \angle AEB = \angle AEC \quad \dots(3) \quad [\text{CPCT}]$$

(iii)

In $\triangle BED$ and $\triangle CED$,

$$BD = CD \quad [\text{Given, } \triangle DBC \text{ is an isosceles triangle}]$$

$$BE = CE \quad [\text{From (2)}]$$

$$DE = DE \quad [\text{Common side}]$$

$\therefore$ By SSS congruence criterion, we get

$$\triangle BED \cong \triangle CED$$

$$\text{Also, } \angle BDE = \angle CDE \quad [\text{CPCT}]$$

$$\therefore \angle BAE = \angle CAE \quad [\text{From (1)}]$$

$$\text{And, } \angle BDE = \angle CDE$$

$\therefore$ AE bisects $\angle A$ as well as $\angle D$.

(iv)

$$BE = CE \quad [\text{From (2)}]$$

$$\therefore \angle AEB + \angle AEC = 180^\circ \quad [\text{Linear pair}]$$

$$\Rightarrow \angle AEB + \angle AEB = 180^\circ \quad [\text{From (3)}]$$

$$\Rightarrow 2\angle AEB = 180^\circ$$

$$\Rightarrow \angle AEB = \frac{180^\circ}{2}$$

$$\Rightarrow \angle AEB = 90^\circ$$

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AE is the perpendicular bisector of BC.

Solution 7:

Given, $x = y$ and $AB = CB$

In $\triangle ABE$,

$$y^\circ = \angle EBA + \angle BAE \quad [\text{Exterior angle property}]$$

In $\triangle BCD$,

$$x^\circ = \angle CBD + \angle BCD \quad [\text{Exterior angle property}]$$

$$\therefore \angle CBD + \angle BCD = \angle EBA + \angle BAE$$

$$\Rightarrow \angle BCD = \angle BAE \quad [\because \angle CBD = \angle EBA]$$

Now,

Consider the triangles AEB and CDB.

$$\angle EBA = \angle DBC$$

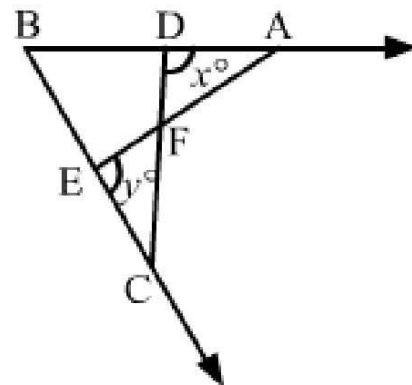
$$\angle BCD = \angle BAE$$

$$AB = CB$$

By ASA congruence criterion, we get

$$\triangle BCD \cong \triangle BAE$$

$$\therefore AE = CD \quad [\text{CPCT}]$$



Solution 8:

Given: $\angle BAQ = \angle BAP$, $BP \perp AP$ and $BQ \perp AQ$.

To prove:

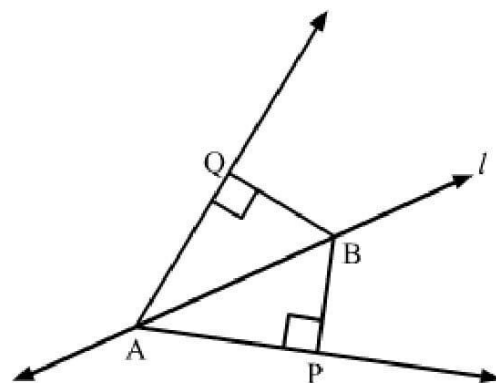
$$(i) \triangle APB \cong \triangle AQB$$

(ii) $BP = BQ$, i.e., B is equidistant from the arms of $\angle A$.

Proof:

(i) In $\triangle APB$ and $\triangle AQB$,

$$\angle BAQ = \angle BAP, \quad [\text{Given}]$$



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$$\angle APB = \angle AQB = 90^\circ \quad [\text{Given,}]$$

$$AB = AB \quad [\text{Common side}]$$

$\therefore$ By AAS congruence criterion, we get

$$\triangle APB \cong \triangle AQB$$

$$(ii) \triangle APB \cong \triangle AQB \quad [\text{Proved above}]$$

$$\therefore BP = BQ \quad [\text{CPCT}]$$

Hence, B is equidistant from the arms of $\angle A$.

Solution 9:

Given: In quadrilateral ABCD, AC bisects the angles $\angle A$ and $\angle C$.

To prove: $AB = AD$ and $CB = CD$

Proof:

In $\triangle ABC$ and $\triangle ADC$,

$$\angle BAC = \angle DAC \quad [\text{Given}]$$

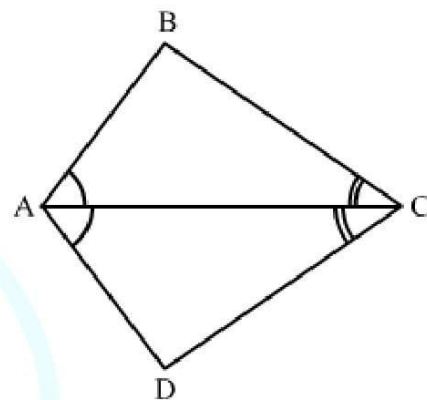
$$AC = AC \quad [\text{Common side}]$$

$$\angle BCA = \angle DCA \quad [\text{Given}]$$

$\therefore$ By ASA congruence criterion, we get

$$\triangle ABC \cong \triangle ADC$$

$$\text{Hence, } AB = AD \text{ and } CB = CD. \quad [\text{CPCT}]$$



Solution 10:

Construction: Draw $DE \perp BC$

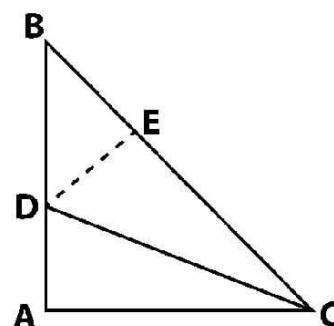
In $\triangle DAC$ and $\triangle DEC$

$$\angle BAC = \angle DAC = 90^\circ \quad [DE \perp BC]$$

$$\angle DCA = \angle DCE \quad [CD \text{ bisect } \angle C]$$

$$CD = CD \quad [\text{Common}]$$

By AAS congruence criterion



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$$\triangle DAC \cong \triangle DEC$$

$$DA = DE \dots (1) \quad \text{[CPCT]}$$

$$AC = EC \dots (2) \quad \text{[CPCT]}$$

It is given that $AB = AC$. So, $\angle B = \angle C$

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow 90^\circ + \angle B + \angle B = 180^\circ$$

$$\Rightarrow 2\angle B = 180^\circ - 90^\circ$$

$$\Rightarrow 2\angle B = 90^\circ$$

$$\Rightarrow \angle B = 45^\circ$$

In $\triangle BED$

$$\angle BED = 90^\circ$$

$$\therefore \angle BDE + \angle B = 90^\circ$$

$$\Rightarrow \angle BDE + 45^\circ = 90^\circ$$

$$\Rightarrow \angle BDE = 90^\circ - 45^\circ$$

$$\Rightarrow \angle BDE = 45^\circ$$

It can be written as

$$\angle BDE = \angle DBE = 45^\circ$$

$$DE = BE \dots (3)$$

From equation (1) and (3)

$$DA = DE = BE \dots (4)$$

Now,

$$BC = BE + EC$$

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$$\Rightarrow BC = DA + AC$$

$$\Rightarrow AC + AD = BC$$

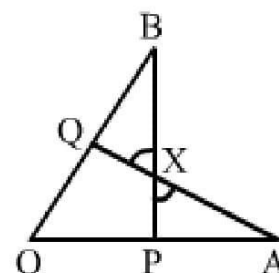
Therefore, it is proved that $AC + AD = BC$.

Solution 11:

Proof:

In $\triangle OQA$ and $\triangle OPB$,

$OQ = OP$	[Given]
$OA = OB$	[Given]
$\angle AOQ = \angle BOP$	[Common]
$\triangle OQA \cong \triangle OPB$	[SAS criterion]
$\angle OAQ = \angle OBP$	[CPCT]



Now, consider triangles BQX and APX .

Given:

$$\begin{aligned} OA &= OB \\ OP &= OQ \\ \therefore OA - OP &= OB - OQ \\ \Rightarrow AP &= BQ \end{aligned}$$

Also,

$\angle BXQ = \angle AXP$	[Vertically opposite angles]
$\angle QBX = \angle PAX$	

By AAS congruence criterion, we get

$\triangle BQX \cong \triangle APX$	[AAS criterion]
$\therefore PX = QX$ and $AX = BX$	[CPCT]

Hence, proved.

Solution 12:

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Given: ABC is an equilateral triangle; $PQ \parallel AC$ and AC is produced to R such that $CR = BP$

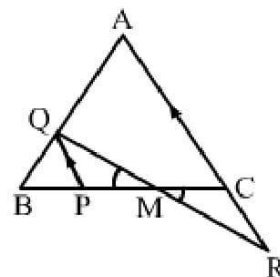
Consider QR intersecting the line PC at point M

From figure,

$$\angle A = \angle ACB = 60^\circ$$

$$\angle BPQ = \angle ACB = 60^\circ$$

[Corresponding angles]



In $\triangle BPQ$

$$\angle B = \angle ACB = 60^\circ$$

$$\angle A = \angle BQP = 60^\circ$$

$\therefore \triangle BPQ$ is an equilateral triangle

$$PQ = BP$$

$$\Rightarrow PQ = CR \text{ (1)}$$

In $\triangle PMQ$ and $\triangle CMR$ we know that $PQ \parallel AC$ and QR is the transversal

We know that $\angle PQM$ and $\angle CRM$ are alternate angles and $\angle PMQ$ and $\angle CMR$ are vertically opposite angles

$$\angle PQM = \angle CRM$$

$$\angle PMQ = \angle CMR$$

By considering equation (1) and AAS congruence criterion

$$\triangle PMQ \cong \triangle CMR$$

We know that the corresponding parts of congruent triangles are equal

$$PM = MC$$

Therefore, it is proved that QR bisects PC.

Solution 13:

Given: ABCD is a quadrilateral in which $AB \parallel DC$ and P is the midpoint of BC.

In $\triangle ABP$ and $\triangle PCQ$

$$\angle PAB = \angle PCQ$$

[Alternate angles]

$$\angle APB = \angle CPQ$$

[Vertically opposite]

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$$PB = PC$$

According to AAS congruence criterion

$$\triangle ABP \cong \triangle PCQ$$

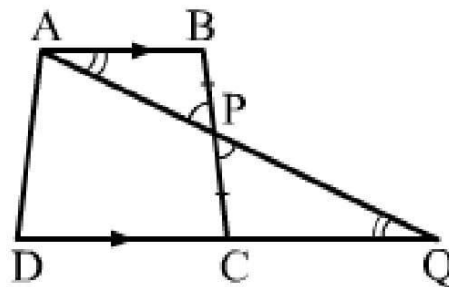
$$AB = CQ \quad \dots(1) \quad \text{[CPCT]}$$

We know that

$$DQ = DC + CQ$$

$$\Rightarrow DQ = DC + AB \quad \text{[From (1)]}$$

Hence, Proved



Solution 14:

In $\triangle APD$ and $\triangle APB$

$$AD = AB \quad \text{[Sides of a square]}$$

$$AP = AP \quad \text{[Common]}$$

$$PD = PB \quad \text{[Given]}$$

By SSS congruence criterion, we get

$$\triangle APD \cong \triangle APB$$

$$\therefore \angle APD = \angle APB \quad \dots(1) \quad \text{[CPCT]}$$

In $\triangle CPD$ and $\triangle CPB$

$$CD = CB \quad \text{[Sides of a square]}$$

$$CP = CP \quad \text{[Common]}$$

$$PD = PB \quad \text{[Given]}$$

By SSS congruence criterion, we get

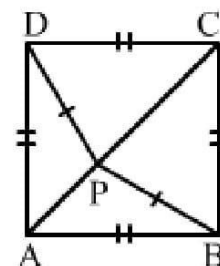
$$\triangle CPD \cong \triangle CPB$$

$$\therefore \angle CPD = \angle CPB \quad \dots(2) \quad \text{[CPCT]}$$

Adding equation (1) and (2),

$$\angle APD + \angle CPD = \angle APB + \angle CPB \quad \dots\dots (3)$$

$$\text{Now, } \angle APD + \angle CPD + \angle APB + \angle CPB = 360^\circ$$



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$$\Rightarrow \angle APB + \angle CPB = 360^\circ - (\angle APD + \angle CPD)$$

$$\Rightarrow \angle APD + \angle CPD = 360^\circ - (\angle APD + \angle CPD) \quad [\text{From equation (3)}]$$

$$\Rightarrow 2(\angle APD + \angle CPD) = 360^\circ$$

$$\Rightarrow \angle APD + \angle CPD = 180^\circ$$

Therefore, CPA is a straight line.

Solution 15:

Since, $\triangle OAB$ is an equilateral triangle

$$\angle OAB = \angle OBA = \angle AOB = 60^\circ$$

Also, ABCD is a square

$$\therefore \angle A = \angle B = \angle C = \angle D = 90^\circ$$

Now,

$$\angle A = \angle DAO + \angle OAB$$

$$\Rightarrow 90^\circ = \angle DAO + 60^\circ$$

$$\Rightarrow \angle DAO = 90^\circ - 60^\circ$$

$$\Rightarrow \angle DAO = 30^\circ$$

$$\text{Also, } \angle CBO = 30^\circ$$

$$\angle DAO = \angle CBO \quad \dots(1)$$

In $\triangle OAD$ and $\triangle OBC$

$$\angle DAO = \angle CBO \quad [\text{From (1)}]$$

$$AD = BC \quad [\text{sides of a square}]$$

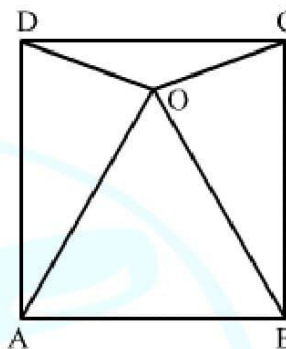
$$OA = OB \quad [\text{sides of an equilateral triangle}]$$

By SAS congruence criterion, we get

$$\triangle OAD \cong \triangle OBC$$

$$\therefore OD = OC \quad [\text{CPCT}]$$

Therefore, that $\triangle OCD$ is an isosceles triangle.

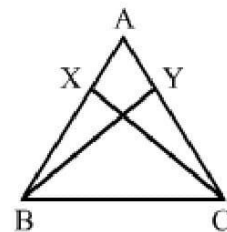


Solution 16:

In $\triangle AXC$ and $\triangle AYB$

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$AX = AY$ [Given]
 $\angle A = \angle A$ [Common]
 $AC = AB$ [Sides of equilateral triangle]
 By SAS congruence criterion, we get
 $\triangle AXC \cong \triangle AYB$
 $\therefore XC = YB$ [CPCT]
 $\Rightarrow CX = BY$.
 Hence, Proved

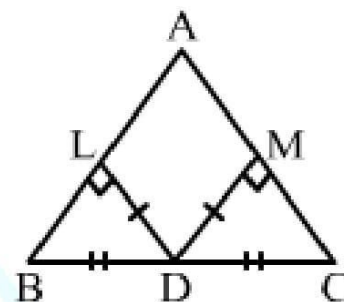


Solution 17:

Given: D is the midpoint of BC
 $DL \perp AB$ and $DM \perp AC$ such that $DL = DM$

In $\triangle BLD$ and $\triangle CMD$

$\angle BLD = \angle CMD$ [$\angle BLD = \angle CMD = 90^\circ$]
 $BD = CD$ [D is midpoint]
 $DL = DM$ [Given]



By RHS congruence criterion, we get

$\triangle BLD \cong \triangle CMD$
 $\therefore \angle ABD = \angle ACD$

Now, in $\triangle ABC$

We know that the sides opposite to equal angles are equal.

Therefore, $AB = AC$.

Solution 18: It is given that $AB = AC$ and the bisectors $\angle B$ and $\angle C$ meet at a point O

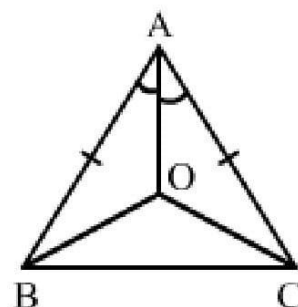
Consider $\triangle BOC$

$\Rightarrow \angle B = \angle C$

$\angle OBC = \frac{1}{2} \angle B$ and $\angle OCB = \frac{1}{2} \angle C$

$\therefore \angle OBC = \angle OCB$

$\Rightarrow OB = OC$... (i)



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$\angle B$ and $\angle C$ has the bisectors OB and OC so we get

$$\angle ABO = \frac{1}{2}\angle B \text{ and } \angle ACO = \frac{1}{2}\angle C$$

So, we get

$$\angle ABO = \angle ACO \dots (ii)$$

In $\triangle ABO$ and $\triangle ACO$ and equation (1) and (2)

$$AB = AC \quad [\text{Given}]$$

$$\angle ABO = \angle ACO \quad [\text{From (ii)}]$$

$$OB = OC \quad [\text{From (i)}]$$

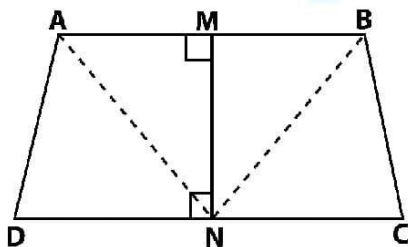
By SAS congruence criterion, we get

$$\triangle ABO \cong \triangle ACO$$

$$\therefore \angle BAO = \angle CAO \quad [\text{CPCT}]$$

Therefore, $BO = CO$ and the ray AO is the bisector of $\angle A$.

Solution 19: Construct AN and BN at the point N



In $\triangle ANM$ and $\triangle BNM$

$$AM = BM \quad [M \text{ is the midpoint of the line } AB]$$

$$\angle AMN = \angle BMN \quad [\text{Each } 90^\circ]$$

$$MN = MN \quad [\text{Common}]$$

By SAS congruence criterion, we have

$$\triangle ANM \cong \triangle BNM$$

$$\therefore AN = BN \quad \dots(i) \quad [\text{CPCT}]$$

$$\angle ANM = \angle BNM \quad [\text{CPCT}]$$

$$\Rightarrow 90^\circ - \angle ANM = 90^\circ - \angle BNM$$

$$\Rightarrow \angle AND = \angle BNC \quad \dots(ii)$$

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Now,

In $\triangle AND$ and $\triangle BNC$

$$AN = BN$$

$$\angle AND = \angle BNC$$

$$DN = CN$$

[N is the midpoint of the line DC]

By SAS congruence criterion,

$$\triangle AND \cong \triangle BNC$$

$$AD = BC$$

[CPCT]

Hence, proved

Solution 20:

In $\triangle ABC$,

It is given that $AB = AC$

$$\therefore \angle ABC = \angle ACB$$

$$\Rightarrow \frac{1}{2} \angle ABC = \frac{1}{2} \angle ACB$$

$$\Rightarrow \angle OBC = \angle OCB$$

[From (i)]

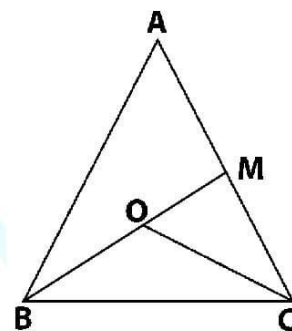
Now,

$$\angle MOC = \angle OBC + \angle OCB \quad [\text{exterior angle property}]$$

$$\Rightarrow \angle MOC = \angle OBC + \angle OBC \quad [\text{from (i)}]$$

$$\Rightarrow \angle MOC = \angle ABC \quad [\text{OB is the bisector of } \angle ABC]$$

Hence, Proved



Solution 21:

In $\triangle ABC$,

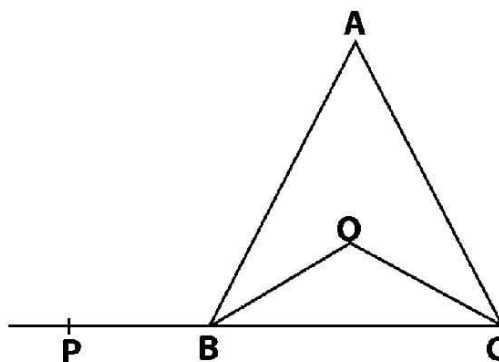
$$AB = AC$$

[Given]

So, we get

$$\angle ABC = \angle ACB$$

$$\Rightarrow \frac{1}{2} \angle ABC = \frac{1}{2} \angle ACB$$



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$$\Rightarrow \angle OBC = \angle OCB \dots\dots (1)$$

Also, In $\triangle BOC$

$$\angle BOC + \angle OBC + \angle OCB = 180^\circ$$

$$\Rightarrow \angle BOC + 2\angle OBC = 180^\circ \quad [\text{From equation (1)}]$$

$$\Rightarrow \angle BOC + \angle ABC = 180^\circ \quad \dots(i)$$

Now,

$$\angle ABC + \angle ABP = 180^\circ \quad [\angle ABC \text{ and } \angle ABP \text{ form a linear pair}]$$

$$\Rightarrow \angle ABC = 180^\circ - \angle ABP \quad \dots(ii)$$

From equation (i) and (ii)

$$\angle BOC + (180^\circ - \angle ABP) = 180^\circ$$

$$\Rightarrow \angle BOC + 180^\circ - \angle ABP = 180^\circ$$

$$\Rightarrow \angle BOC - \angle ABP = 0$$

$$\Rightarrow \angle BOC = \angle ABP$$

Therefore, it is proved that the exterior angle adjacent to $\angle ABC$ is equal to $\angle BOC$.

Solution 22:

We know that $AB \parallel PQ$ and BP is the transversal

From the figure, we have

$$\angle ABP = \angle BPQ \dots(1) \quad [\text{Alternate angles}]$$

Since, BP is the bisector of $\angle ABC$

$$\therefore \angle ABP = \angle PBC \text{ and } \angle ABP = \angle PBQ \dots (2)$$

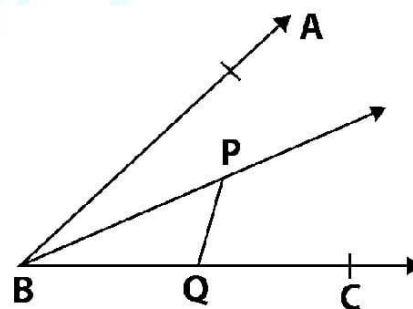
From equation (1) and (2), we get

$$\angle BPQ = \angle PBQ$$

We know that the sides opposite to equal angles are equal

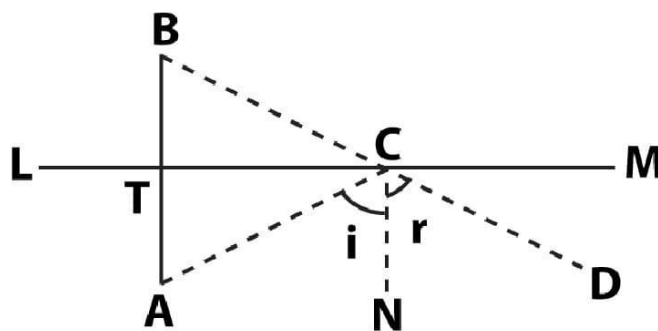
$$PQ = BQ$$

Therefore, it is proved that $\triangle BPQ$ is an isosceles triangle.



Solution 23:

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Given: An object is placed at a point A, the image of the object is seen at the point B, an observer is at point D, and LM is a plane mirror.

To Prove: The image is as far behind the mirror as the object is in front of the mirror, i.e., $BT = AT$.

Angle of incidence = Angle of reflection

$$\angle ACN = \angle DCN \quad \dots (i)$$

We have $AB \parallel CN$ and AC is the transversal

$$\angle TAC = \angle ACN \quad \dots (ii) \quad [\text{Alternate interior angles}]$$

Also, $AB \parallel CN$ and BD is the transversal

$$\angle TBC = \angle DCN \quad \dots (iii) \quad [\text{Corresponding angles}]$$

From the equation (i), (ii) and (iii), we get

$$\angle TAC = \angle TBC \quad \dots (iv)$$

Now,

In $\triangle ACT$ and $\triangle BCT$

$$\angle ATC = \angle BTC \quad [\text{Each of } 90^\circ]$$

$$CT = CT \quad [\text{Common}]$$

$$\angle TAC = \angle TBC \quad [\text{From (iv)}]$$

By AAS congruence criterion, we get

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$$\triangle ACT \cong \triangle BCT$$

$$AT = BT$$

Therefore, it is proved that the image is as far behind the mirror as the object is in front of the mirror.

Solution 24:

Let AB be the breadth of the river.

Take a point M at a distance from B.

Let O be the midpoint of BM.

Draw a perpendicular from the point M and name it as N so that it joins the point A as a straight line.

Now,

In $\triangle ABO$ and $\triangle NMO$

$$\angle OBA = \angle OMN \quad [\text{Each of } 90^\circ]$$

$$OB = OM \quad [O \text{ is the midpoint of the line } BM]$$

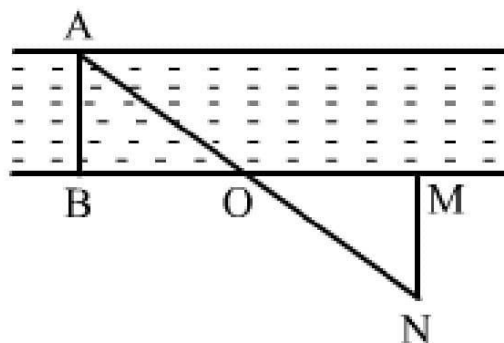
$$\angle BOA = \angle MON \quad [\text{Vertically opposite angles}]$$

By ASA congruence criterion

$$\triangle ABO \cong \triangle NMO$$

$$AB = NM \quad [\text{CPCT}]$$

Therefore, MN is the width of the river.



Solution 25:

We have, D is the midpoint of the line AC

$$\therefore AD = CD = \frac{1}{2} AC$$

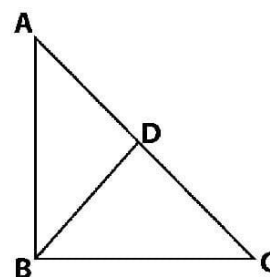
$$\Rightarrow AD = CD = BD \quad [\because BD = \frac{1}{2} AC]$$

$$\Rightarrow \angle BAD = \angle ABD \quad \dots(i) \quad [\because AD = BD, \text{ angles opposite to equal sides are equal}]$$

Also,

$$\angle BCD = \angle CBD \quad \dots(ii) \quad [\because CD = BD, \text{ angles opposite to equal sides are equal}]$$

Now, In $\triangle ABC$



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$$\angle ABC + \angle BAC + \angle BCA = 180^\circ$$

$$\Rightarrow \angle ABC + \angle BAD + \angle BCD = 180^\circ$$

By using equation (i) and (ii), we get

$$\Rightarrow \angle ABC + \angle ABD + \angle CBD = 180^\circ$$

$$\Rightarrow \angle ABC + \angle ABC = 180^\circ$$

$$\Rightarrow 2\angle ABC = 180^\circ$$

$$\Rightarrow \angle ABC = 90^\circ$$

Therefore, $\angle ABC$ is a right angle.

Solution 26: No, if two sides and the included angle of one triangle are equal to corresponding two sides and the included angle of another triangle then the two triangles are congruent. Otherwise, the triangles will not congruent to each other.

Solution 27: Yes, the statement is true because the two triangles can be congruent by either AAS or ASA congruence criteria.

Study Path

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EXERCISE 9B

Solution 1:

(i) No, because the sum of two sides of a triangle is not greater than the third side.
i.e., $5 + 4 = 9$ is not greater than 9.

(ii) Yes, because the sum of two sides of a triangle is greater than the third side.
 $7 + 4 > 8$; $8 + 7 > 4$; $8 + 4 > 7$

(iii) Yes, because the sum of two sides of a triangle is greater than the third side.
 $5 + 6 > 10$; $10 + 6 > 5$; $5 + 10 > 6$

(iv) Yes, because the sum of two sides of a triangle is greater than the third side.
 $2.5 + 5 > 7$; $5 + 7 > 2.5$; $2.5 + 7 > 5$

(v) No, because the sum of two sides of a triangle is not greater than the third side.
i.e., $3 + 4 = 7$ is not greater than 8

Solution 2:

Given: In $\triangle ABC$, $\angle A = 50^\circ$ and $\angle B = 60^\circ$

In $\triangle ABC$,

$$\angle A + \angle B + \angle C = 180^\circ \quad (\text{Angle sum property of a triangle})$$

$$\Rightarrow 50^\circ + 60^\circ + \angle C = 180^\circ$$

$$\Rightarrow 110^\circ + \angle C = 180^\circ$$

$$\Rightarrow \angle C = 180^\circ - 110^\circ$$

$$\Rightarrow \angle C = 70^\circ$$

$$\therefore \angle A < \angle B < \angle C$$

$$BC < AC < AB$$

Hence, the longest side will be opposite to the largest angle ($\angle C = 70^\circ$) i.e. AB.

And, the shortest side will be opposite to the smallest angle ($\angle A = 50^\circ$) i.e. BC.

Solution 3:

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(i) Given: In $\triangle ABC$, $\angle A = 90^\circ$

$$\therefore \angle B + \angle C = 90^\circ$$

$$\text{i.e. } \angle B, \angle C < 90^\circ$$

Since, $\angle A$ is the greatest angle.

So, the longest side is BC.

(ii) Given:

$$\angle A = \angle B = 45^\circ$$

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow 45^\circ + 45^\circ + \angle C = 180^\circ$$

$$\Rightarrow \angle C = 90^\circ$$

Since, $\angle C$ is the greatest angle.

So, the longest side is AB.

(iii) Given:

$$\angle A = 100^\circ \text{ and } \angle C = 50^\circ$$

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow 100^\circ + \angle B + 50^\circ = 180^\circ$$

$$\Rightarrow \angle B = 30^\circ$$

Since, $\angle A$ is the greatest angle.

So, the shortest side is AC.

Solution 4:

In triangle CBA, CBD is an exterior angle.

$$\text{i.e., } \angle CBA + \angle CBD = 180^\circ$$

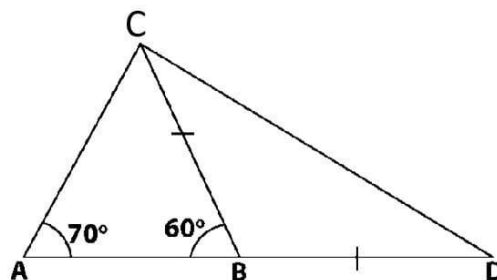
$$\Rightarrow 60^\circ + \angle CBD = 180^\circ$$

$$\Rightarrow \angle CBD = 120^\circ$$

Triangle BCD is isosceles and $BC = BD$.

$$\therefore \angle BCD = \angle BDC$$

In $\triangle CBD$, we have:



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$$\Rightarrow \angle BCD + \angle CBD + \angle CDB = 180^\circ$$

$$\Rightarrow \angle BCD + 120^\circ + \angle BCD = 180^\circ$$

$$\Rightarrow 2\angle BCD = 60^\circ$$

$$\Rightarrow \angle BCD = 30^\circ$$

$$\therefore \angle BCD = \angle BDC = 30^\circ$$

In $\triangle ADC$,

$$\angle C = \angle ACB + \angle BCD = 50^\circ + 30^\circ = 80^\circ$$

$$\angle A = 70^\circ \text{ and } \angle D = 30^\circ$$

$$\therefore \angle C > \angle A$$

$$\Rightarrow AD > CD \quad \dots(1)$$

$$\text{Also, } \angle C > \angle D$$

$$\Rightarrow AD > AC \quad \dots(2)$$

Solution 5:

Given: $\angle B < \angle A$ and $\angle C < \angle D$

To prove: $AD < BC$

Proof:

In $\triangle AOB$,

$$\angle B < \angle A$$

$$\Rightarrow AO < BO \quad \dots(1) \quad (\text{Side opposite to the greater angle is longer})$$

In $\triangle COD$,

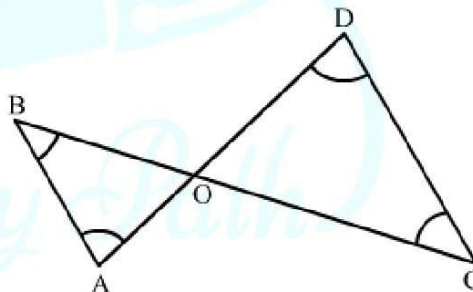
$$\angle C < \angle D$$

$$\Rightarrow OD < OC \quad \dots(2) \quad (\text{Side opposite to the greater angle is longer})$$

Adding (1) and (2), we get

$$AO + OD < BO + OC$$

$$\therefore AD < BC$$



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Solution 6: Given: In quadrilateral ABCD, AB and CD are respectively the smallest and largest sides.

To prove:

- (i) $\angle A > \angle C$
- (ii) $\angle B > \angle D$

Construction: Join AC.

Proof:

In $\triangle ABC$,

$BC > AB$ (Given, AB is the smallest side)

$$\therefore \angle BAC > \angle ACB \quad \dots(1)$$

In $\triangle ADC$,

$CD > AD$ (Given, CD is the largest side)

$$\therefore \angle CAD > \angle ACD \quad \dots(2)$$

Adding (1) and (2), we get

$$\angle BAC + \angle CAD > \angle ACD + \angle ACB$$

$$\therefore \angle A > \angle C$$

Construction: Join BD.

Proof:

In $\triangle ABD$,

$AD > AB$ (Given, AB is the smallest side.)

$$\therefore \angle ABD > \angle ADB \quad \dots(3)$$

In $\triangle CBD$,

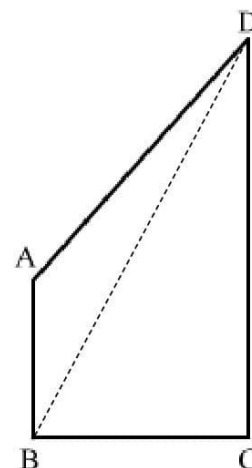
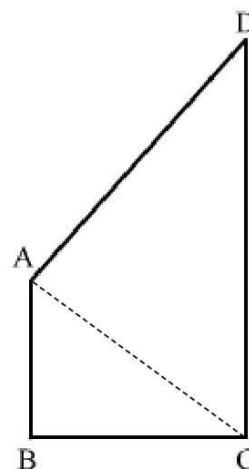
$CD > BC$ (Given, CD is the greatest side.)

$$\therefore \angle DBC > \angle CDB \quad \dots(4)$$

Adding (3) and (4), we get

$$\angle ABD + \angle DBC > \angle ADB + \angle CDB$$

$$\therefore \angle B > \angle D$$



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Solution 7:

Given: Quadrilateral ABCD

To prove: $(AB + BC + CD + DA) > (AC + BD)$

Proof:

In $\triangle ABC$,

$$AB + BC > AC \quad \dots(i)$$

In $\triangle CAD$,

$$CD + AD > AC \quad \dots(ii)$$

In $\triangle BAD$,

$$AB + AD > BD \quad \dots(iii)$$

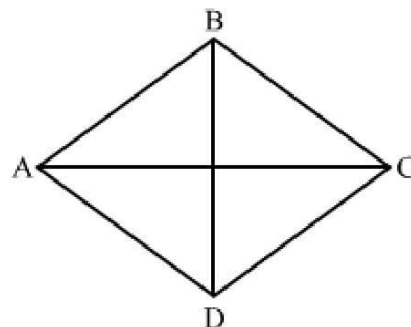
In $\triangle BCD$,

$$BC + CD > BD \quad \dots(iv)$$

Adding (i), (ii), (iii) and (iv), we get

$$2(AB + BC + CD + DA) > 2(AC + BD)$$

$$\therefore (AB + BC + CD + DA) > (AC + BD).$$



Solution 8: Given: Quadrilateral ABCD

To prove: $(AB + BC + CD + DA) < 2(BD + AC)$.

Proof:

In $\triangle AOB$,

$$OA + OB > AB \quad \dots(i)$$

In $\triangle BOC$,

$$OB + OC > BC \quad \dots(ii)$$

In $\triangle COD$,

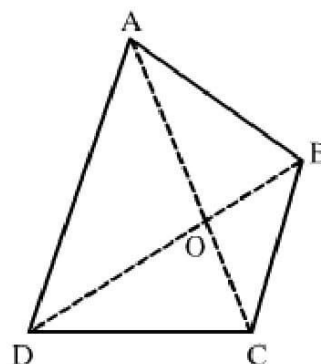
$$OC + OD > CD \quad \dots(iii)$$

In $\triangle AOD$,

$$OD + OA > AD \quad \dots(iv)$$

Adding (i), (ii), (iii) and (iv), we get

$$2(OA + OB + OC + OD) > (AB + BC + CD + DA)$$



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$$\Rightarrow 2(OB + OD + OA + OC) > (AB + BC + CD + DA)$$

$$\Rightarrow 2(BD + AC) > (AB + BC + CD + DA)$$

$$\Rightarrow (AB + BC + CD + DA) < 2(BD + AC)$$

Solution 9: Given: In $\triangle ABC$, $\angle B = 35^\circ$, $\angle C = 65^\circ$ and the bisector of $\angle BAC$ meets BC in X .

In $\triangle ABX$,

$$\angle BAX > \angle ABX$$

$$\therefore BX > AX \quad \dots(i)$$

Similarly,

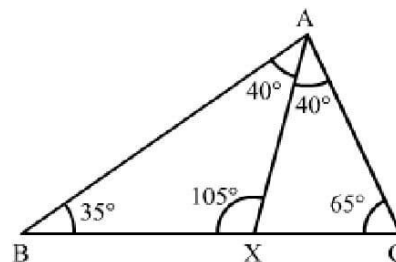
In $\triangle ACX$,

$$\angle ACX > \angle XAC$$

$$\therefore AX > CX \quad \dots(ii)$$

From (i) and (ii), we get

$$BX > AX > CX$$



Solution 10:

Since the angle opposite to the longer side is greater

$$\therefore PQ > PR$$

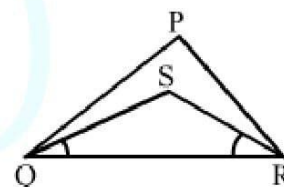
$$\Rightarrow \angle R > \angle Q$$

$$\Rightarrow \frac{1}{2}\angle R > \frac{1}{2}\angle Q$$

$$\Rightarrow \angle SRQ > \angle RQS$$

$$\Rightarrow QS > SR$$

$$\therefore SQ > SR$$



Solution 11:

Given: In $\triangle ABC$, $AB = AC$

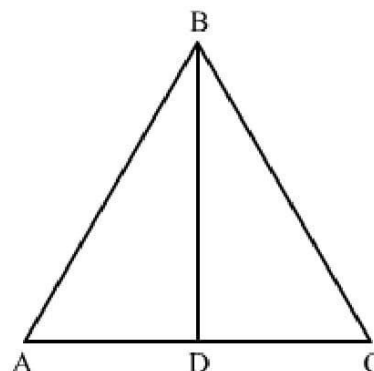
To prove: $CD < BD$

Proof:

In $\triangle ABC$,

$$AB = AC$$

[Given]



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So, $\angle ABC = \angle ACB$... (1)

In $\triangle ABC$ and $\triangle DBC$,

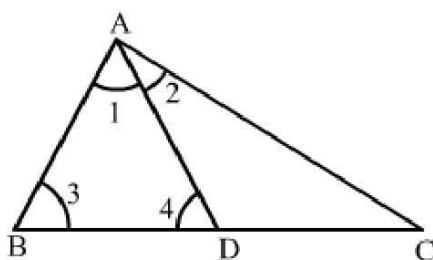
$\angle ABC > \angle DBC$

$\Rightarrow \angle ACB > \angle DBC$ [From (1)]

$\Rightarrow BD > CD$ [Side opposite to greater angle is longer]

$\therefore CD < BD$

Solution 12:



Given: In $\triangle ABC$, BC is the longest side.

To prove: $\angle BAC > \frac{2}{3} \times 90^\circ = 60^\circ$

Construct: Mark a point D on side AC such that $AD = AB = BD$.

Proof:

In $\triangle ABD$,

$\therefore AD = AB = BD$ (By construction)

$\therefore \angle 1 = \angle 3 = \angle 4 = 60^\circ$

Now,

$\angle BAC = \angle 1 + \angle 2 = 60^\circ + \angle 2$

$\angle BAC = \frac{2}{3}$ of a right angle + $\angle 2$

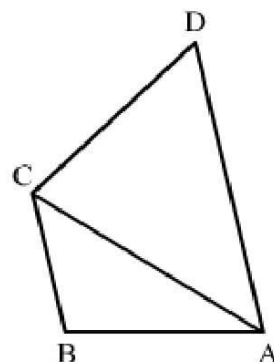
Hence, $\angle BAC > \frac{2}{3}$ of a right angle.

Solution 13:

Given: Quadrilateral ABCD

To prove:

(i) $CD + DA + AB > BC$



(ii) $CD + DA + AB + BC > 2AC$

Proof:

(i) In $\triangle ACD$,

$$CD + DA > CA \quad \dots(1)$$

In $\triangle ABC$,

$$AB + CA > BC \quad \dots(2)$$

Adding (1) and (2), we get

$$CD + DA + AB + CA > CA + BC$$

$$\therefore AB + CD + DA > BC$$

(ii) In $\triangle CDA$,

$$CD + DA > CA \quad \dots(3)$$

In $\triangle BCA$,

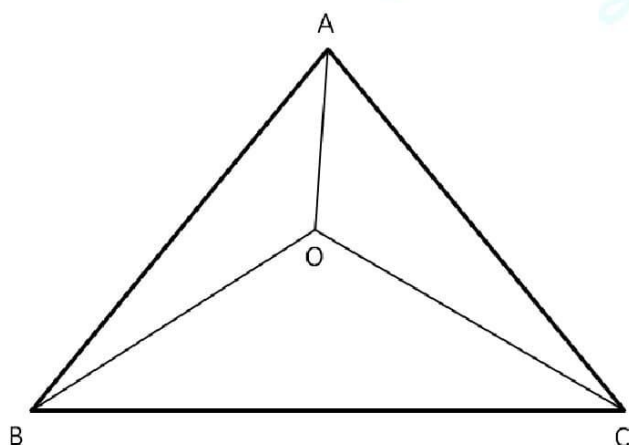
$$BC + AB > CA \quad \dots(4)$$

Adding (3) and (4), we get

$$CD + AD + BC + AB > CA + CA$$

$$\therefore CD + AD + BC + AB > 2CA$$

Solution 14:



(i) It is given that O is a point within $\triangle ABC$

In $\triangle ABC$,

$$AB + AC > BC \quad \dots (1)$$

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In $\triangle OBC$,

$$OB + OC > BC \dots (2)$$

Subtracting (2) from equation (1), we get

$$(AB + AC) - (OB + OC) > BC - BC$$

$$\Rightarrow (AB + AC) - (OB + OC) > 0$$

$$\Rightarrow AB + AC > OB + OC$$

Therefore, it is proved that $AB + AC > OB + OC$.

(ii) We know that

$$AB + AC > OB + OC \dots (1)$$

Also,

$$AB + BC > OA + OC \dots (2)$$

$$AC + BC > OA + OB \dots (3)$$

Adding equation (1), (2) and (3), we get

$$AB + AC + AB + BC + AC + BC > OB + OC + OA + OC + OA + OB$$

$$\Rightarrow 2(AB + BC + AC) > 2(OA + OB + OC)$$

$$\Rightarrow AB + BC + AC > OA + OB + OC$$

(iii) In $\triangle OAB$

$$OA + OB > AB \dots (1)$$

In $\triangle OBC$

$$OB + OC > BC \dots (2)$$

In $\triangle OCA$

$$OC + OA > CA \dots (3)$$

By adding equation (1), (2) and (3), we get

$$OA + OB + OB + OC + OC + OA > AB + BC + CA$$

$$\Rightarrow 2(OA + OB + OC) > AB + BC + CA$$

$$\Rightarrow OA + OB + OC > \frac{1}{2}(AB + BC + CA)$$

Hence, proved

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Solution 15:

Given: $AD \perp BC$ and $CD > BD$

To prove: $AC > AB$

Proof:

$$\angle ADB = \angle ADC = 90^\circ \quad (AD \perp BC) \quad \dots(1)$$

$$\angle BAD < \angle DAC \quad (CD > BD) \quad \dots(2)$$

In $\triangle ABD$

$$\angle B = 180^\circ - \angle ADB - \angle BAD$$

$$\angle B = 90^\circ - \angle BAD \quad \dots(3)$$

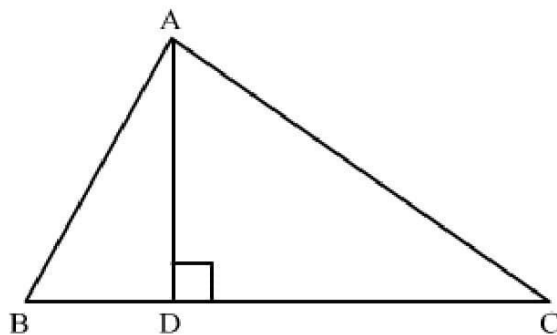
In $\triangle ADC$

$$\angle C = 90^\circ - \angle DAC \quad \dots(4)$$

From (2), (3) and (4), we get

$$\angle B > \angle C$$

$$\therefore AC > AB.$$



Solution 16:

Given: $CD = DE$

To prove: $AB + AC > BE$

Proof:

In $\triangle ABC$,

$$AB + AC > BC \quad \dots(1)$$

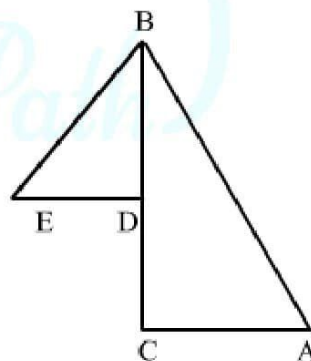
In $\triangle BED$,

$$BD + CD > BE$$

$$\Rightarrow BC > BE \quad \dots(2)$$

From (1) and (2), we get

$$AB + AC > BE$$



Multiple Choice Questions (MCQ)

Solution 1: (a) SSA

Solution 2: (c) $\triangle CAB \cong \triangle PQR$

$$AB = QR, BC = RP, CA = PQ$$

$$\therefore \triangle CAB \cong \triangle PQR$$

Solution 3: (a) $BC = PQ$

Solution 4: (c) 80°

In $\triangle ABC$,

Since, $AB = AC$ so, $\triangle ABC$ is an isosceles triangle,

$$\therefore \angle B = \angle C = 50^\circ$$

Now,

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow \angle A + 50 + 50 = 180^\circ$$

$$\Rightarrow \angle A = 180^\circ - 100^\circ$$

$$\therefore \angle A = 80^\circ$$

Solution 5: (a) 50°

In $\triangle ABC$, $BC = AB$ and $\angle B = 80^\circ$.

$\therefore \triangle ABC$, is isosceles triangle and $\angle C = \angle A$

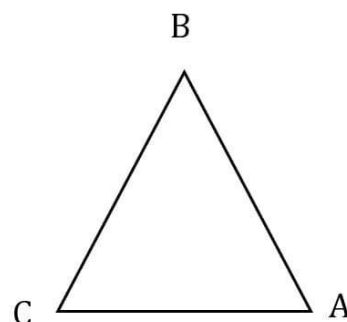
$$\angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow \angle A + 80^\circ + \angle A = 180^\circ$$

$$\Rightarrow 2\angle A = 180^\circ - 80^\circ$$

$$\Rightarrow 2\angle A = 100^\circ$$

$$\Rightarrow \angle A = 50^\circ$$



Solution 6: (a) 4 cm

In $\triangle ABC$, $\angle C = \angle A$, $BC = 4$ cm and $AC = 5$ cm.

$\therefore \triangle ABC$ is isosceles triangle and $AB = BC$

$$\Rightarrow BC = AB = 4$$

Hence, the correct option is (a).

Solution 7: (b) 6.5 cm

The sum of any two sides of a triangle is greater than the third side.

$$\text{Since, } 4 \text{ cm} + 2.5 \text{ cm} = 6.5 \text{ cm}$$

The length of third side of a triangle cannot be 6.5 cm.

Solution 8: (b) $AB > AC$

In $\triangle ABC$,

$$\angle C > \angle B$$

The side opposite to the greater angle is larger.

$$\therefore AB > AC$$

Solution 9: (b) $\angle E = 60^\circ$

$$\triangle ABC \cong \triangle FDE$$

$$AB = 5 \text{ cm}, \angle B = 40^\circ, \angle A = 80^\circ \text{ and } FD = 5 \text{ cm}$$

$$\text{Then } \angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow 80^\circ + 40^\circ + \angle C = 180^\circ$$

$$\Rightarrow \angle C = 60^\circ$$

$$\text{Also, } \angle C = \angle E = 60^\circ$$

Solution 10: (c) AB

In $\triangle ABC$,

$$\angle A = 40^\circ, \angle B = 60^\circ$$

Now,

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow 60^\circ + 40^\circ + \angle C = 180^\circ$$

$$\Rightarrow \angle C = 80^\circ$$

Now,

$$40^\circ < 60^\circ < 80^\circ$$

$$\angle A < \angle B < \angle C$$

i.e., $\angle C$ is greater angle and $\angle A$ is smaller angle.

Now,

$$\angle A < \angle B < \angle C$$

$$BC < AC < AB$$

Since, side opposite to greater angle is larger and side opposite to smaller angle is smaller.

$\therefore$ AB is longest and BC is shortest side.

Solution 11: (c) $AB > AD$

$$\therefore AB > AC$$

$$\therefore \angle ACB > \angle ABC$$

$$\text{Now, } \angle ADB > \angle ACD \quad [\text{exterior angle}]$$

$$\Rightarrow \angle ADB > \angle ACB > \angle ABC$$

$$\Rightarrow \angle ADB > \angle ABD$$

$$\Rightarrow AB > AD$$

Solution 12: (c) $OB > OC$

$$\therefore AB > AC$$

$$\Rightarrow \angle C > \angle B$$

$$\Rightarrow \angle C > \angle B$$

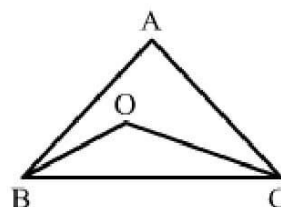
$$\Rightarrow \frac{1}{2} \angle C > \frac{1}{2} \angle B$$

$$\Rightarrow \frac{1}{2} \angle C > \frac{1}{2} \angle B$$

$$\Rightarrow \angle OCB > \angle OBC$$

$$\Rightarrow \angle OCB > \angle OBC$$

$$\Rightarrow OB > OC$$



$\Rightarrow OB > OC$

Solution 13: (a) 1:1

In $\triangle OAB$ and $\triangle OAC$,

$\triangle OAB$ and $\triangle OAC$,

$AB = AC$ [Given]

$OB = OC$ [Given]

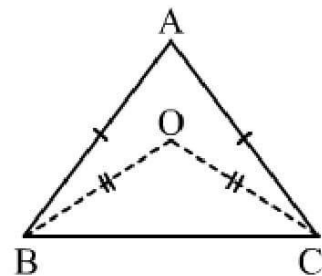
$OA = OA$ [Common]

By SSS congruence criterion, we get

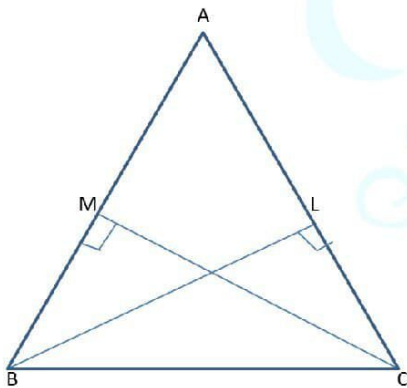
$\triangle OAB \cong \triangle OAC$

$\therefore \angle ABO = \angle ACO$ [CPCT]

$\therefore \angle ABO : \angle ACO = 1 : 1$



Solution 14: (b) isosceles



In $\triangle ABC$, $BL \perp AC$.

$CM \perp AB$ such that $BL = CM$.

To prove: $AB = AC$

In $\triangle ABL$ and $\triangle ACM$,

$BL = CM$ [Given]

$\angle BAL = \angle CAM$ [Common angle]

$\angle ALB = \angle AMC$ [Each 90°]

By AAS congruence criterion, we get

$$\triangle ABL \cong \triangle ACM$$

$$\therefore AB = AC$$

[CPCT]

Solution 15: (b) $\angle B = \angle E$

In $\triangle ABC$ and $\triangle DEF$,

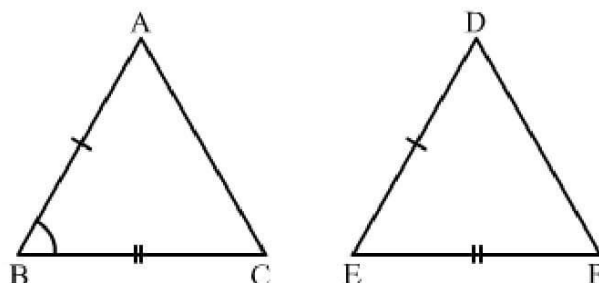
$$AB = DE$$

[Given]

$$BC = EF$$

[Given]

In order that $\triangle ABC \cong \triangle DEF$, we must have $\angle B = \angle E$.



Solution 16: (c) $BC = EF$

Solution 17: (a) isosceles but not congruent

Solution 18: (c) A triangle can have two acute angles.

Solution 19:

- (a) Sum of any two sides of a triangle $>$ the third side
- (b) Difference of any two sides of a triangle $<$ the third side
- (c) Sum of three altitudes of a triangle $<$ sum of its three side
- (d) Sum of any two sides of a triangle $>$ twice the median to the 3rd side
- (e) Perimeter of a triangle $>$ sum of its three medians

Solution 20:

- (a) 60°
- (b) equal.
- (c) longest
- (d) not possible.